Technical Documentation – Project Progress

Project Name: MANDOOBI

Trainee Name: Hatem Salem, Ziad Abdelhamid, Reem ElMoghazy

Date: 20/8/2025

1. Summary

Developed a comprehensive platform using **Angular**, **Firebase**, and **Node.js** with features including product browsing (with dedicated pages for Home, Products, and Special Offers), shopping cart management, user authentication (email/password & phone OTP), search with filtering/sorting, and a checkout process. The backend Express server integrates the Firebase Admin SDK for secure user registration and authentication.

2. Tasks Completed

Task	Description	Status
Home Page	Implemented main image banner, categories section, featured offers section with product cards (including package size selection & quantity controls)	Done
Products Page	Integrated Firestore product list with filters (brand & type selection), price range slider, and sorting options (name/price); search queries (from navbar) are debounced and applied to results	Done
Cart Page	Built cart view with item list, quantity update, remove item, and summary; supports guest cart (stored in localStorage) and syncs with Firestore for logged-in users; includes address input, delivery options, and payment method selection for checkout	Done
Offers Page	Created a dedicated offers page (/offers) listing all special promotional products (using a static OFFERS dataset) with similar card UI and an embedded promotional video banner	Done
Authentication	Developed login & registration (email/password) via Firebase Auth; implemented phone number OTP login via a custom Node.js Express API using Firebase Admin SDK (serving as a fallback since full web OTP is limited)	Done
Navbar & Search	Implemented a responsive navbar with dynamic menu links (Home, Products, Offers, Cart); integrated a search bar that uses **RxJS debounce** and navigates to the Products page with query parameters to filter product results live	Done
Backend Server	Created an Express.js backend (server.js) that uses Firebase Admin SDK to handle secure operations (user creation for registration, issuing custom tokens for login, initiating phone OTP verification); the server is separate from Angular and runs on Node.js	Done
Footer and Banners	Added a responsive footer with quick links and contact info; implemented multiple banner components (image carousel, offer banners, and video banners) on pages for promotions (e.g., an embedded video ad on Home/Offers) to enhance UX	Done

3. Technical Details

- **Frontend:** Angular (standalone components, Angular 17+), TypeScript, HTML5, SCSS for styling.
- **Backend:** Node.js with Express.js server, integrated with Firebase Admin SDK.
- **Firebase Services:** Firestore (for product catalog, cart data, user profiles), Firebase Authentication (for email/password and phone auth).
- **Libraries & Tools:** AngularFire (Angular Firebase integration), RxJS (for reactive programming like search debounce and state management), Bootstrap (for quick styling & responsive layout, as needed).
- **APIs/Data:** The product data is stored in Firestore (retrieved via AngularFire). The Offers page uses a static in-app dataset for special offers. Authentication APIs (login, register, phone OTP) are exposed via the custom Express server which communicates with Firebase.

4. Implementation Steps

Explain the steps you followed to implement your part of the project:

1. **Project Setup:** Initialized a new Angular project and structured it by feature (using standalone components for pages and shared components). Set up Firebase project and integrated AngularFire for Firestore and Auth.

2. **Routing Configuration:** Defined routes in app.routes.ts for Home , Products , Offers , and Cart pages, each route pointing to its respective component.

3. **Home Page:** Built the HomeComponent with a main image carousel banner, a CategoriesComponent (displaying product categories), a VideoBannerComponent for an autoplay promotional video, and a featured offers section. The featured offers section pulls a subset of products from a static OFFERS array (imported from OffersComponent) to highlight daily deals, allowing users to select package sizes (e.g., 6, 12, 24 cans) and adjust quantity.

4. **Products Page:** Developed the ProductsComponent to display all products from Firestore. Implemented reactive search and filtering: the navbar search query is passed via route query params (q) and triggers a Firestore query (with startAt/endAt for prefix matching). Also added brand and type filters (toggled via checkboxes), a price range filter, and client-side sorting (by name or price, ascending/descending). Used RxJS combineLatest to merge all filter observables and produce a filtered, sorted product list in real-time.
   Technical detail: The Products page uses collectionData from AngularFire to get products and applies filters in an RxJS pipeline. A loading indicator and error handling are included for better UX.

5. **Offers Page:** Created an OffersComponent that displays all promotional products (the OFFERS dataset). This page is similar to a product listing but for special deals. It reuses the product card UI for each offer (with image, price, package options,

quantity, and an "Add to Cart" button). A VideoBanner (or a special offers banner) is embedded here to showcase a promotional video or graphic at the top or bottom of the offers list.

6. **Cart & Checkout:** Built the CartComponent and a CartService to manage cart state. For guest users (not logged in), the cart is kept in localStorage. For authenticated users, cart items are stored in Firestore under the user's profile. On login, if a guest user had items, the CartService merges local items into the Firestore cart. The Cart page shows items with quantity controls and allows removal. It also includes a section for the user to select/add an address (for delivery), choose a delivery method (if multiple options), and select a payment method (currently just placeholder options since no payment gateway integration yet). On clicking "Checkout", it simulates order placement (e.g., could save an order record or simply clear the cart and show a confirmation message).

7. **Authentication (Login/Registration + Phone OTP):** Implemented an AuthService and LoginComponent for handling user auth. The LoginComponent is designed as a modal dialog that can switch between Login and Register forms, and also handle Phone OTP flow. For email/password, it directly uses Firebase Auth (via AngularFireAuth). For phone OTP, due to Firebase JS SDK limitations on the web (needing reCAPTCHA, etc.), a custom Express endpoint /initiate-phone-auth was created. This endpoint, when given a phone number, uses Firebase Admin SDK to send a verification code (OTP) to the phone. The user then enters the code in the app, which is verified via another Express endpoint or by signing in with a custom token. Authentication state is managed via currentUser BehaviorSubject in AuthService, and persisted with localStorage so that refreshes maintain the login state.

8. **Navbar & Search:** Built a NavbarComponent that displays the app name/logo, navigation links (Home, Products, Offers, Cart), and shows a login button or user name depending on auth state. The navbar also contains a search input; keystrokes are handled with RxJS debounceTime to avoid excessive queries, and on submit or sufficient typing, it navigates to /products?q=searchTerm. The Cart icon in the navbar includes a badge showing the number of items in the cart (which the CartService keeps track of).

9. **Footer and Banners:** Created a FooterComponent with company info, links, and social icons to appear at the bottom of all pages. Developed multiple banner components for visual appeal:

    - *MainBannerComponent:* an image carousel on the Home page with promotional slides.

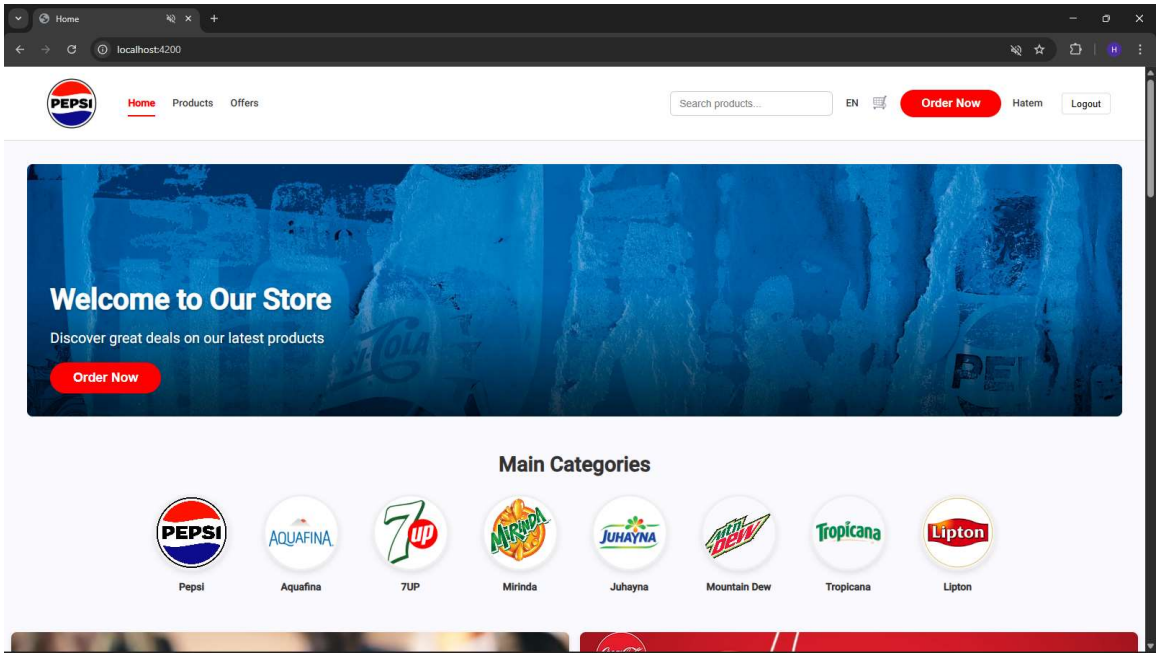
    - *OffersBannerComponent:* a banner highlighting special deals (used on Offers or Home page).

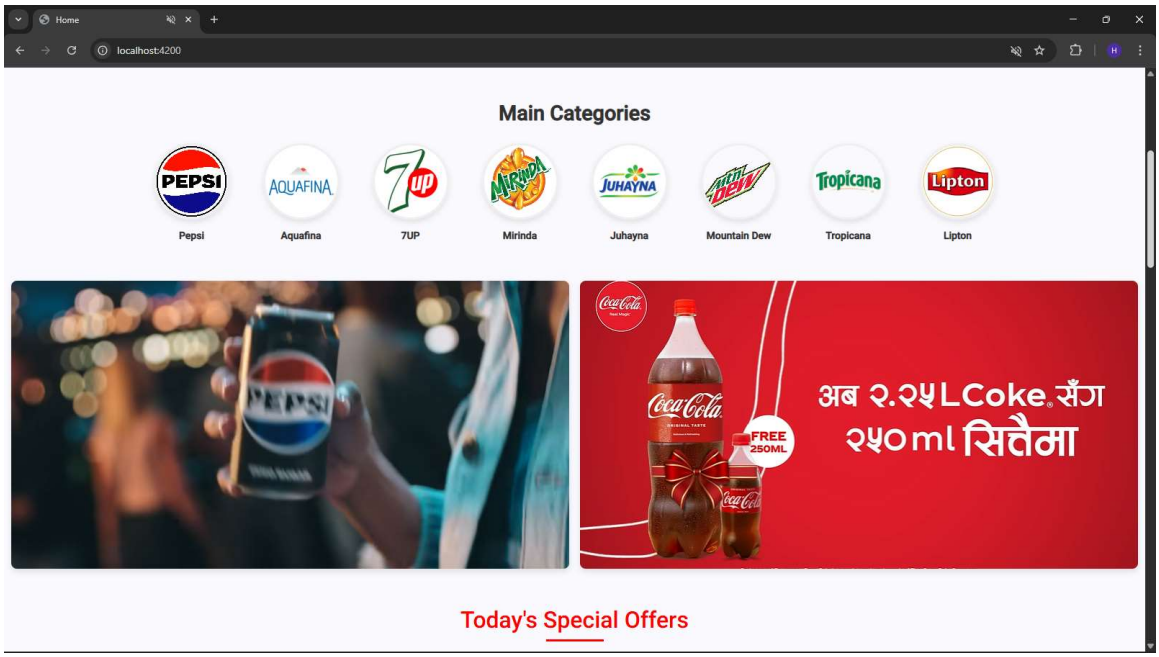
- *VideoBannerComponent & Vid2BannerComponent:* components to play promotional videos on loop (used on Home/Offers page for dynamic effect). These components are all standalone and included in the Home/Offers pages to make the UI lively and marketing-friendly.

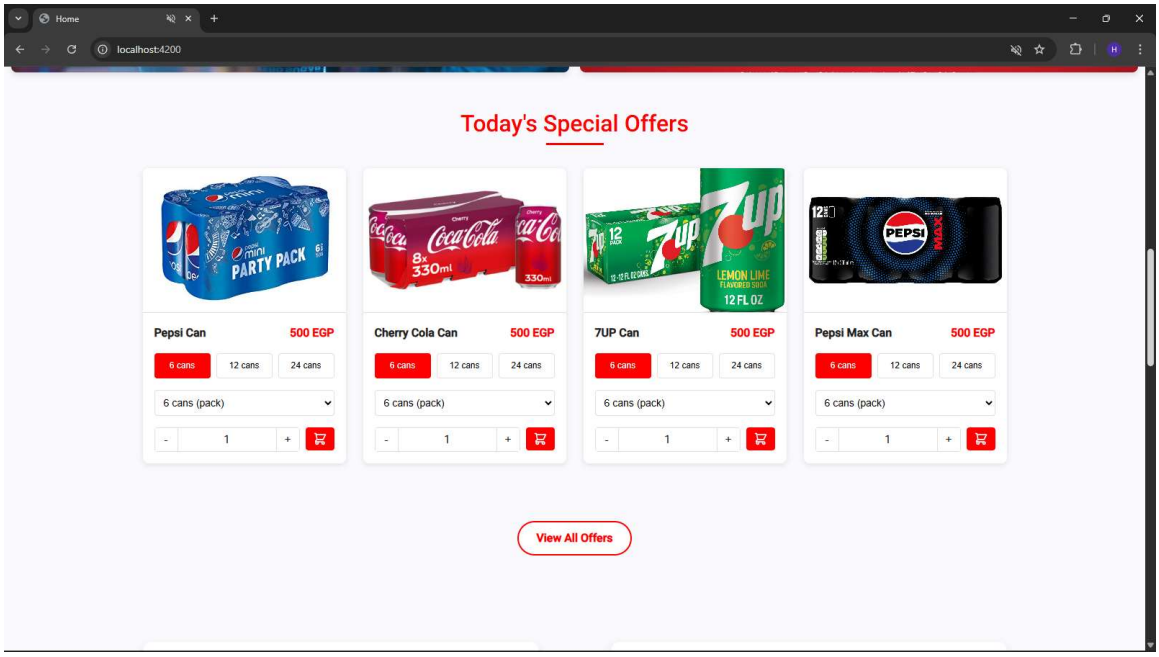
10. **Backend Express Server:** Set up a simple Express server (backend/server.js) to handle sensitive operations. This server initializes the Firebase Admin SDK with a service account to securely perform admin tasks. Implemented routes such as:

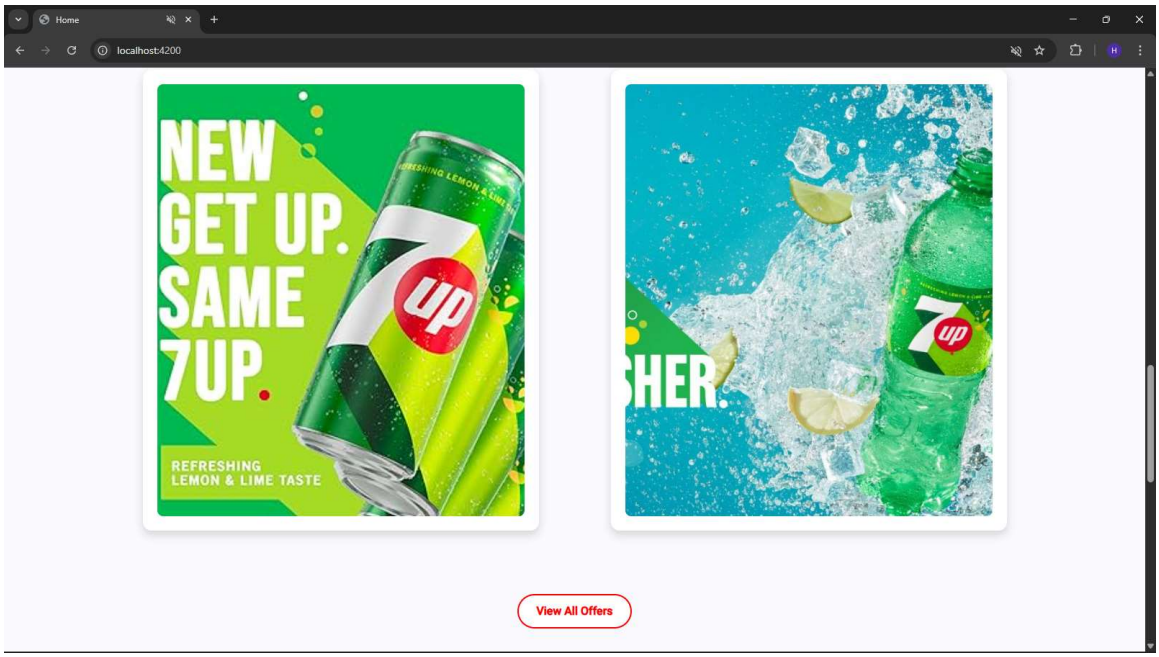
    - POST /register – create a new user (Firebase Auth) with email and password.
    - POST /login – verify user credentials and issue a custom auth token.
    - POST /initiate-phone-auth – use Admin SDK to send an OTP to a given phone number.
    - POST /verify-phone-code (if applicable) – verify the OTP and return a Firebase custom token.
      These endpoints allow the Angular app to perform registration and phone authentication securely (without exposing Admin credentials on the client). CORS is enabled so the Angular dev server can call these APIs. The Angular environment config contains the URL to this backend (e.g., http://localhost:3000).

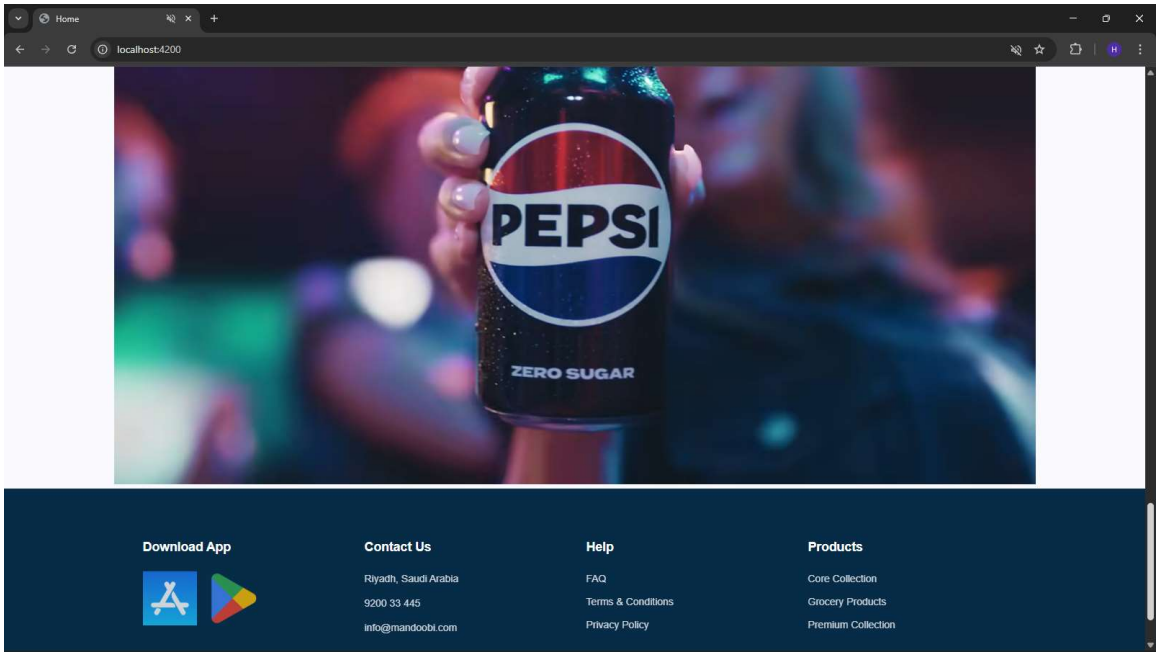
6. Screenshots / Code Snippets

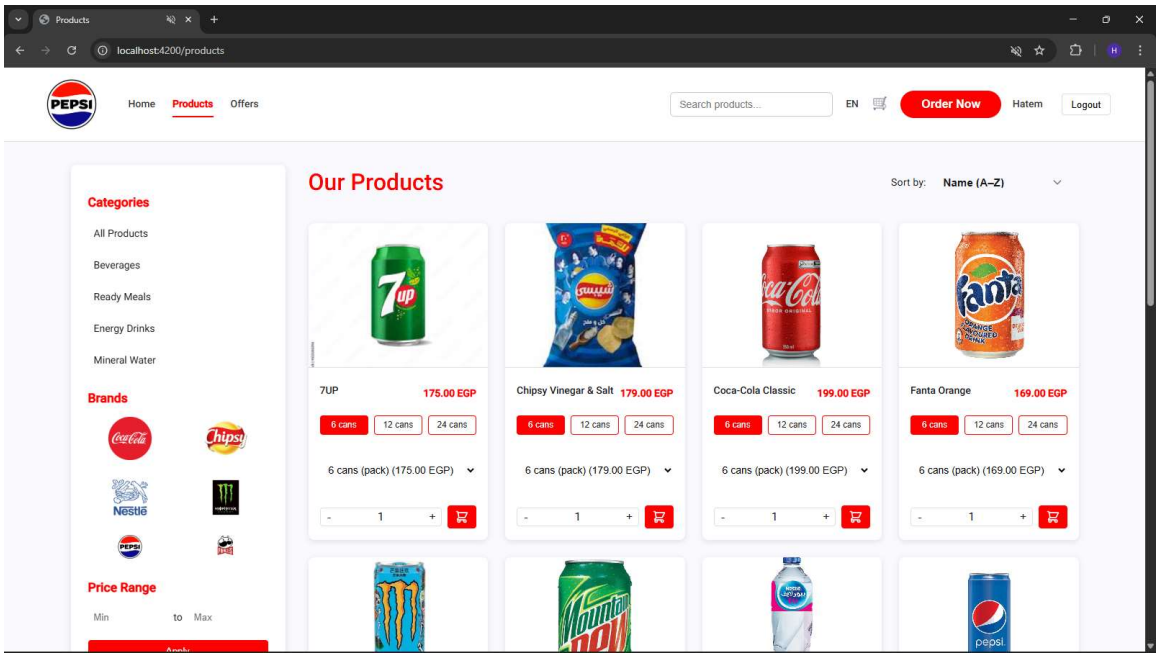


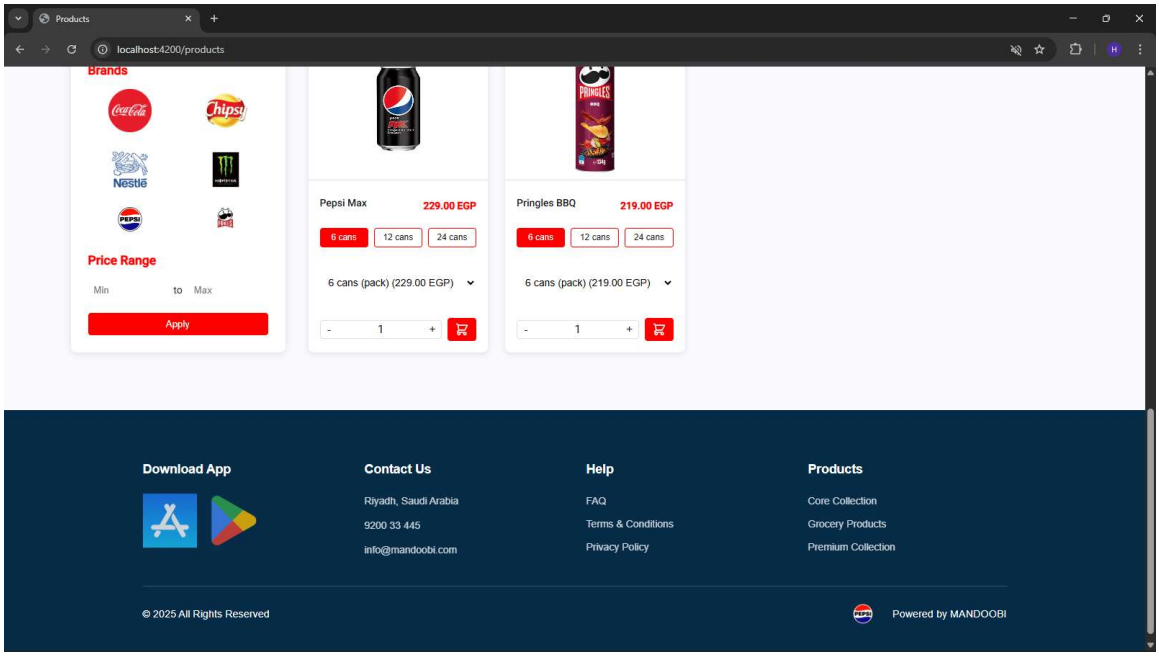

Home
localhost:4200
Main Categories
Pepsi
Aquafina
7UP
Mirinda
Juhayna
Mountain Dew
Tropicana
Lipton
अब २.२५L Coke सँग
२५० ml सित्तैमा
FREE 250ML
Today's Special Offers

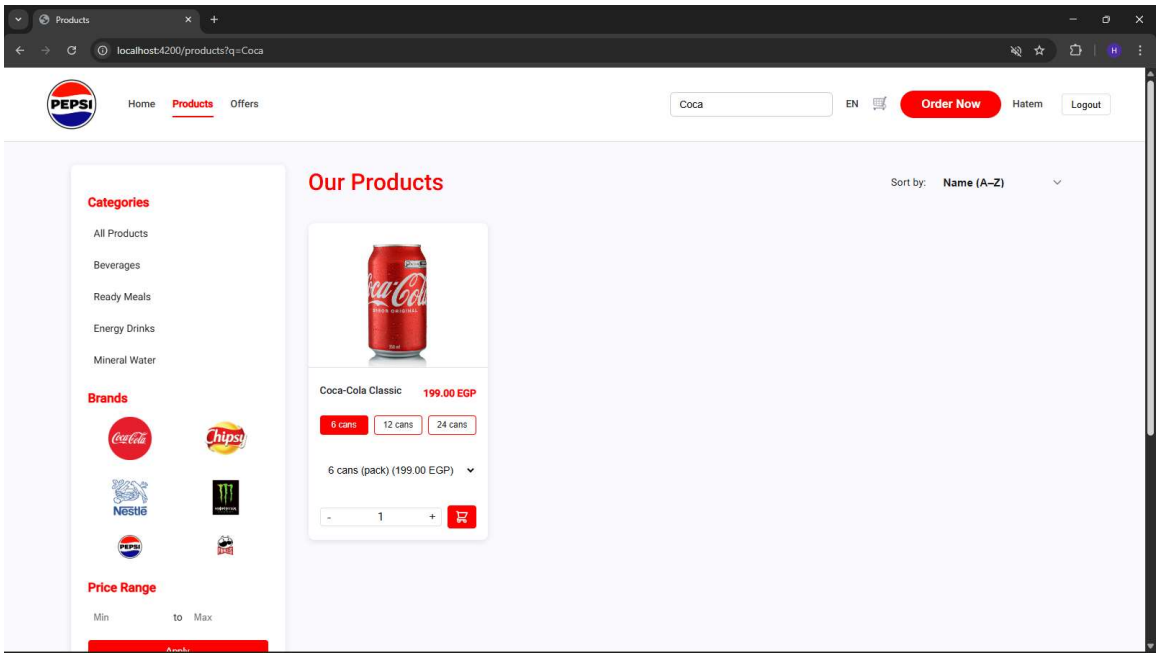

Home
localhost:4200
Today's Special Offers
Pepsi Can
500 EGP
Cherry Cola Can
500 EGP
7UP Can
500 EGP
Pepsi Max Can
500 EGP
6 cans
12 cans
24 cans
6 cans (pack)
View All Offers

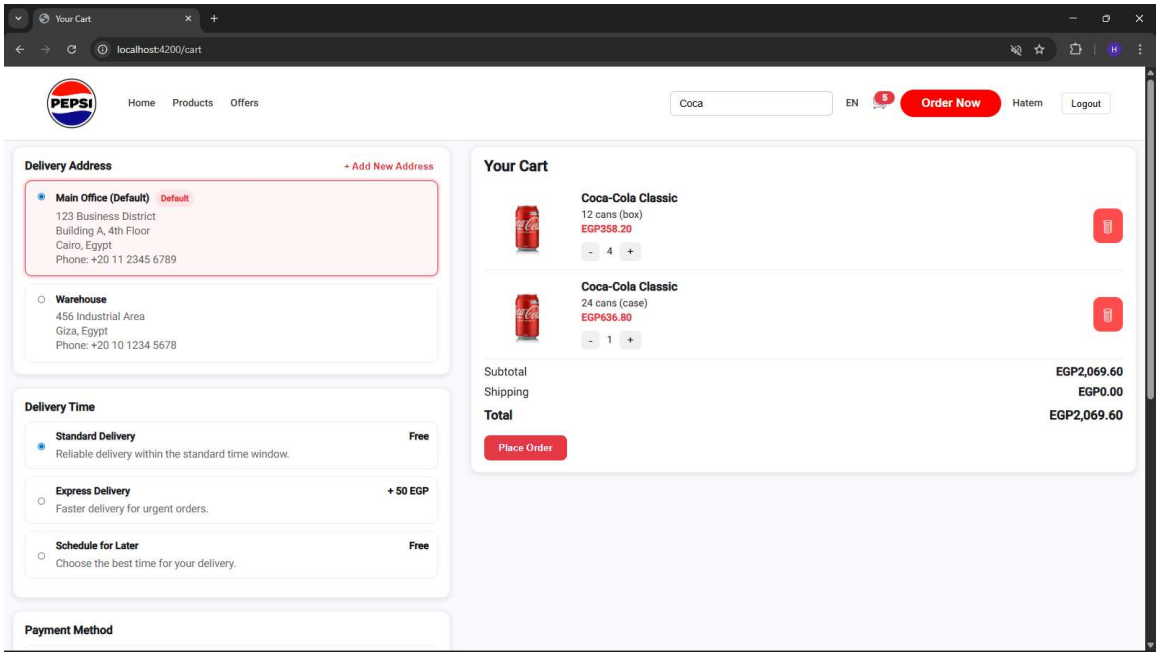

Home
localhost:4200
NEW
GET UP.
SAME
7UP.
REFRESHING
LEMON & LIME TASTE
View All Offers

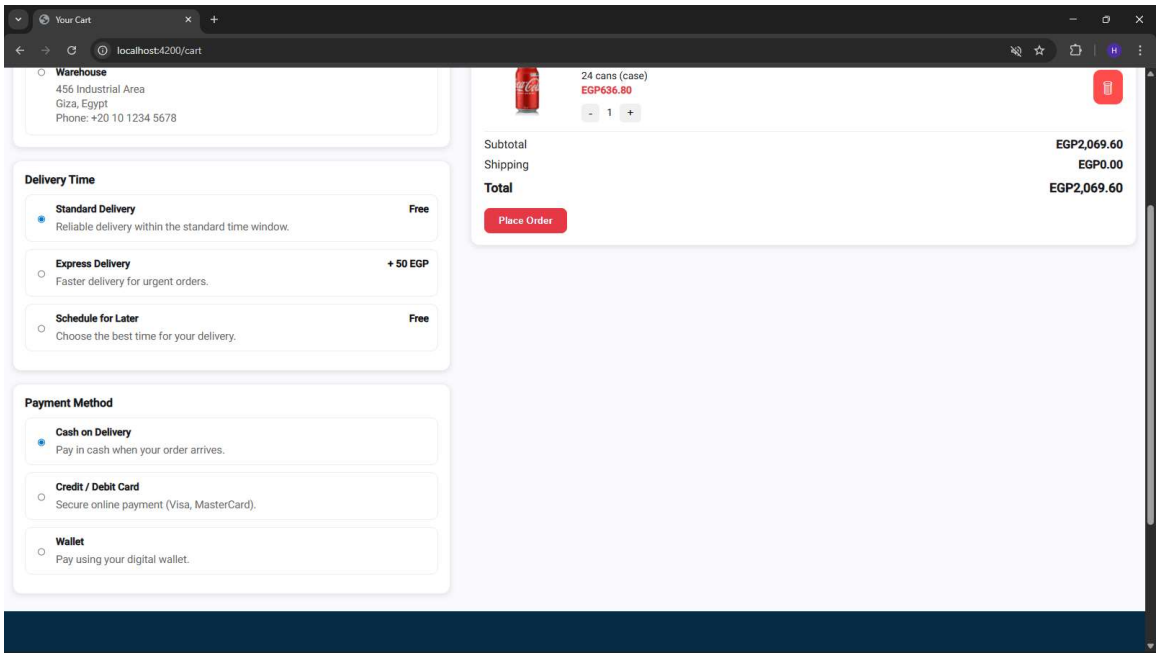

Home
localhost:4200
PEPSI
ZERO SUGAR
Download App
Contact Us
Riyadh, Saudi Arabia
9200 33 445
info@mandoobi.com
Help
FAQ
Terms & Conditions
Privacy Policy
Products
Core Collection
Grocery Products
Premium Collection

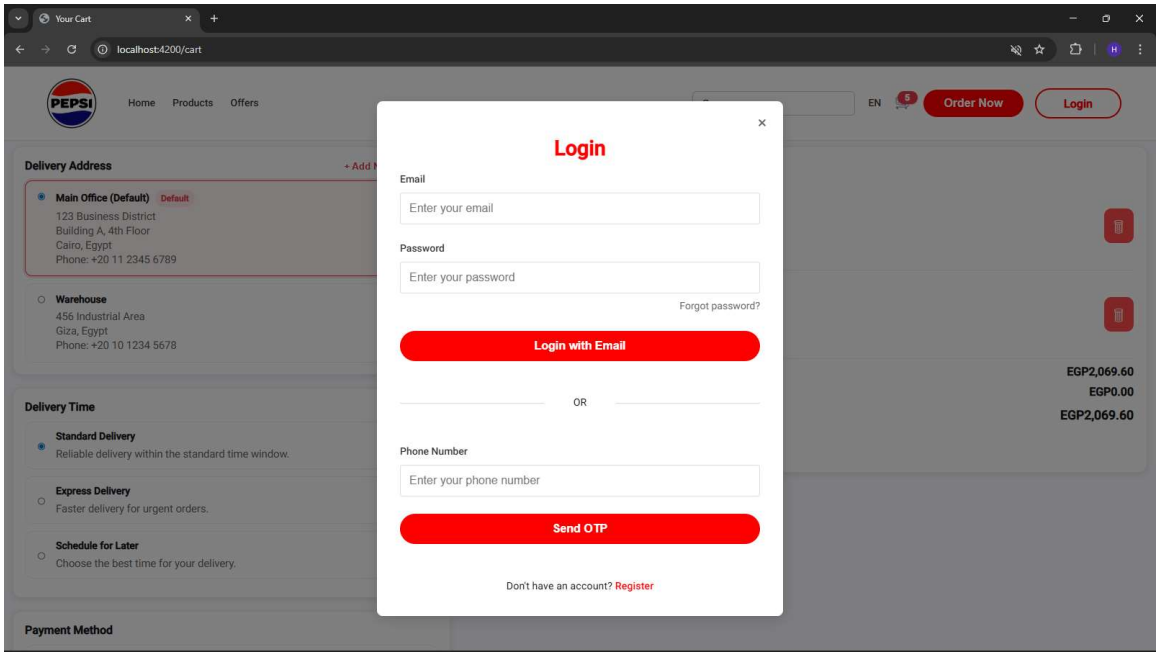

Products
localhost:4200/products
PEPSI
Home
Products
Offers
Search products...
EN
Order Now
Hatem
Logout
Categories
All Products
Beverages
Ready Meals
Energy Drinks
Mineral Water
Brands
Price Range
Min
to
Max
Our Products
Sort by:
Name (A–Z)
7UP
175.00 EGP
Chipsy Vinegar & Salt
179.00 EGP
Coca-Cola Classic
199.00 EGP
Fanta Orange
169.00 EGP
6 cans
12 cans
24 cans
6 cans (pack) (175.00 EGP)
6 cans (pack) (179.00 EGP)
6 cans (pack) (199.00 EGP)
6 cans (pack) (169.00 EGP)

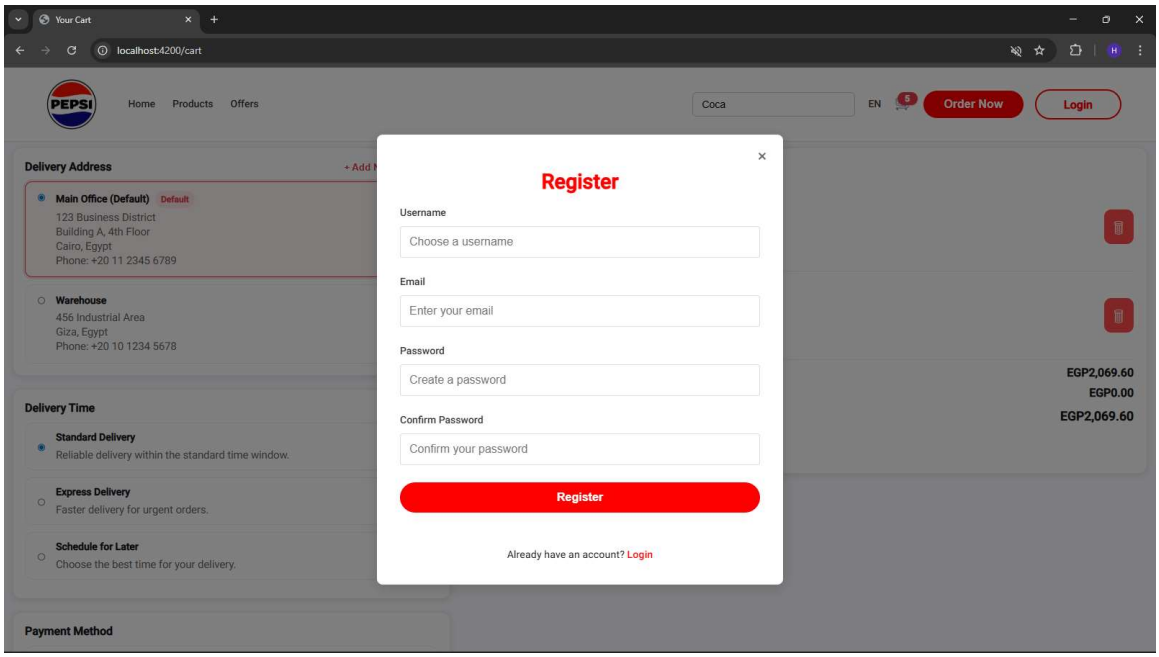

Products
localhost:4200/products
Brands
Price Range
Min
to
Max
Apply
Pepsi Max
229.00 EGP
Pringles BBQ
219.00 EGP
6 cans
12 cans
24 cans
6 cans (pack) (229.00 EGP)
6 cans (pack) (219.00 EGP)
Download App
Contact Us
Riyadh, Saudi Arabia
9200 33 445
info@mandoobi.com
Help
FAQ
Terms & Conditions
Privacy Policy
Products
Core Collection
Grocery Products
Premium Collection
© 2025 All Rights Reserved
Powered by MANDOOBI


Products
localhost:4200/products?q=Coca
PEPSI
Home
Products
Offers
Coca
EN
Order Now
Hatem
Logout
Categories
All Products
Beverages
Ready Meals
Energy Drinks
Mineral Water
Brands
Price Range
Min
to
Max
Our Products
Sort by:
Name (A–Z)
Coca-Cola Classic
199.00 EGP
6 cans
12 cans
24 cans
6 cans (pack) (199.00 EGP)


Your Cart
localhost:4200/cart
PEPSI
Home
Products
Offers
Coca
EN
Order Now
Hatem
Logout
Delivery Address
+ Add New Address
Main Office (Default)
Default
123 Business District
Building A, 4th Floor
Cairo, Egypt
Phone: +20 11 2345 6789
Warehouse
456 Industrial Area
Giza, Egypt
Phone: +20 10 1234 5678
Delivery Time
Standard Delivery
Free
Reliable delivery within the standard time window.
Express Delivery
+ 50 EGP
Faster delivery for urgent orders.
Schedule for Later
Free
Choose the best time for your delivery.
Payment Method
Your Cart
Coca-Cola Classic
12 cans (box)
EGP358.20
Coca-Cola Classic
24 cans (case)
EGP636.80
Subtotal
EGP2,069.60
Shipping
EGP0.00
Total
EGP2,069.60
Place Order


Your Cart
localhost:4200/cart
Warehouse
456 Industrial Area
Giza, Egypt
Phone: +20 10 1234 5678
Delivery Time
Standard Delivery
Free
Reliable delivery within the standard time window.
Express Delivery
+ 50 EGP
Faster delivery for urgent orders.
Schedule for Later
Free
Choose the best time for your delivery.
Payment Method
Cash on Delivery
Pay in cash when your order arrives.
Credit / Debit Card
Secure online payment (Visa, MasterCard).
Wallet
Pay using your digital wallet.
24 cans (case)
EGP636.80
- 1 +
Subtotal
EGP2,069.60
Shipping
EGP0.00
Total
EGP2,069.60
Place Order


Your Cart
localhost:4200/cart
PEPSI
Home
Products
Offers
EN
Order Now
Login
Delivery Address
Main Office (Default)
Default
123 Business District
Building A, 4th Floor
Cairo, Egypt
Phone: +20 11 2345 6789
Warehouse
456 Industrial Area
Giza, Egypt
Phone: +20 10 1234 5678
Delivery Time
Standard Delivery
Reliable delivery within the standard time window.
Express Delivery
Faster delivery for urgent orders.
Schedule for Later
Choose the best time for your delivery.
Payment Method
EGP2,069.60
EGP0.00
EGP2,069.60
Login
Email
Enter your email
Password
Enter your password
Forgot password?
Login with Email
OR
Phone Number
Enter your phone number
Send OTP
Don't have an account? Register



5. Challenges Faced & Future Solutions

Challenge	Solution
Phone OTP Authentication Not Working: The current phone login flow only checks if a phone number exists in Firebase via the backend. If the number is not found, the user is redirected to the register page. No OTP SMS is ever sent because the Firebase Admin SDK cannot send OTPs, and the frontend does not use the required signInWithPhoneNumber + reCAPTCHA flow.	Integrate proper OTP support by adding signInWithPhoneNumber and RecaptchaVerifier on the Angular client side to request OTPs, while keeping the backend (server.js) only for token issuance. This will allow Firebase to send OTPs to the user's phone and enable a complete login flow.
Search Performance: Current implementation fetches all products and applies client-side filtering. When the search box is blank, previously filtered results may remain visible.	Refactor ProductsComponent to maintain two arrays: allProducts (everything from Firestore) and filteredProducts (UI list). Use RxJS (debounceTime, distinctUntilChanged) to minimize queries. If search is blank → reset filteredProducts = allProducts; if not blank → apply filter.
Responsive Design & Multiple Banners: Multiple banners, product grids, and categories section may not display consistently across mobile and desktop.	Tested each page on different screen sizes, adjusting breakpoints for the carousel and video banners. The categories section on Home collapses into a carousel on mobile. The layout and spacing of product cards and offers were tuned to prevent overflow. This ensured a consistent and responsive design throughout.

Guest Cart vs User Cart Synchronization: Guest carts (localStorage) and user carts (Firestore) risk duplication or item loss when a user logs in.	Enhance CartService merge logic: compare items, sum quantities for duplicates, and resolve conflicts by prioritizing server-side data. After syncing, clear the guest cart. Add test cases to ensure consistent behavior across login transitions.

7. Next Steps / To-Do

- **Order History & Management:** Implement an Order Confirmation page after checkout and an Order History page where logged-in users can see past orders (stored in Firestore).
- **Product Detail Page:** Add a dedicated product details page that opens when a user clicks a product from the Products or Offers list, showing a larger image, detailed description, reviews, etc.
- **Payment Gateway Integration:** Integrate a real payment gateway (e.g., Apple Pay or PayPal) to handle actual payments on checkout, replacing the current simulated payment flow.
- **Enhanced UI/UX:** Add a wishlist feature for users to save items for later, a reviews/ratings system for products, and perhaps a dynamic carousel for the main banner. Continue refining responsive design, especially for the Offers and Products grid on smaller screens.
- **Admin Module:** Develop an admin dashboard (could be a separate Angular app or added routes) for managing products, offers, and viewing orders. This would use Firebase security rules or additional backend APIs to restrict access.